

INDIA'S DEEPTech MOMENT



Indian deep-tech start-ups raised \$1.3 bn in 2024, up 78% YoY. Top 10 deals totalled \$600 million



India produces over 1.5 mn engineering graduates annually, making it one of the largest STEM talent pools globally



Returnee founders (ex-NASA, SpaceX, Intel, Qualcomm) bringing global expertise home



300+ specialised research labs, IITs, IISc, and DRDO fuelling deeptech entrepreneurship



India saw over 90,000 patents filed in FY24, marking the 7th consecutive year of growth at 11.3%



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encourage start-ups to own IP-designed products.”

ISM is short for India Space Mission and DLI stands for Design-Linked Incentive scheme.

FROM SOFTWARE TO PHYSICAL TECHNOLOGY

The transition becomes particularly interesting when viewed across sectors.

At Exponent Energy, co-founder and CEO Arun Vinayak says the company’s starting point was not to build another electric vehicle but to solve what he sees as the harder problem—energy.

Commercial vehicles account for only 10% of vehicles but consume 70% of India’s energy, he says. Exponent has built a technology stack around rapid charging and battery durability. Its batteries last

3,000-4,000 cycles and charge in 15 minutes; the company says they are designed to last one million kilometres.

“For the first time, large OEMs (original equipment manufacturers) now have an Indian technology partner, not just a supplier for plastic or sheet metal parts,” says Vinayak.

That distinction—technology partner rather than component supplier—could define this new phase of Indian manufacturing.

The problem is not simply inventing something. It is about building everything around the invention. That becomes even more apparent in aerospace.

ROCKETS, AIRCRAFT, CHIPS

Astrobase co-founder and CEO

Neeraj Khandelwal believes reusability will eventually transform the economics of launch vehicles.

“In 10-15 years, every rocket will need to be reusable—landing back and flying again instead of building a new one each time,” he says. “Reusability changes the entire economic equation.”

The company’s proposition is also part of a much larger space transformation in India, which is targeting a space economy of more than \$40 billion by 2040.

The important change is that private companies are no longer simply supplying components to India’s space programme. They are attempting to build launch vehicles, engines, satellite systems and downstream applications.

“ISRO has done an amazing job for 50-60 years, but it cannot do this alone,” says Khandelwal. “The space sector needs to be industrialised, and private companies are very good at industrialising things.”

E-Plane represents another version of the same ambition.

The company is currently on the Directorate General of Civil Aviation (DGCA) certification pathway for its electric aircraft. It expects Phase II to be completed by October, followed by prototype construction and flight testing. It is targeting the end of the first quarter of 2028 for full certification.

EXPENSIVE FINAL GAP

Professor Satya Chakravarty, the company’s founder, says the underlying technology is 100% indigenous. The 20% of parts that are currently imported account for about 50% of the cost, he says.

It is a revealing statistic because it captures India’s deep-tech challenge in one number—the